RDS Project - Final Version

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Proposed ADS: Home Credit Default Risk
Competition Host: Home Credit Group
Link: https://www.kaggle.com/competitions/home-credit-default-risk
Solution: https://github.com/abhishekdbihani/Home-Credit-Default-Risk-Recognition

We selected this ADS, because home loans are crucial to economic mobility, and whether a person is approved or denied can dramatically change the trajectory of their lives. It's therefore important to promote both group fairness and individual fairness, so a marginalized group isn't left behind, and everyone who deserves a loan gets it. Since there is historical bias in this field we want to ensure that the models we create are fair and uphold principles of equity.

Background

The ADS that we plan to analyze in this project is a submission into Home Credit Group's Home Credit Default Risk Kaggle competition. Home Credit Group is a lender that strives to provide loans to underserved populations. The goal of the Kaggle competition was to develop a classifier tool that can assist Home Credit Group in their decision-making process for providing loans. The model we decided to audit classifies applicants as high-risk of default vs low. We decided to research the fairness of the model when it comes to sex. Historically, males tend to default more on their loans than females, so we'd like to see how the model handles these differences in base rates. In this audit we will prioritize a group fairness framework to ensure similar outcomes among the two groups.

Credit history is an important factor assessed by lenders when determining the trustworthiness of prospective borrowers. For many, credit serves as a lifeline when money is not readily available. Through this ADS, the submitter hopes to utilize historical data to develop a tool that can be used for future credit lending decision-making. One of the major trade-offs that this model could yield is an inclusion of potentially high-risk borrowers. This may introduce a higher-likelihood of defaults on loans. This conflicts with the goal of providing credit to underserved groups.

Input and output

The data used was sourced from Home Credit Group for the Kaggle competition. The data collected was that of previous applicants (historical data). They extensively represent applicant information. The data files contain various data points that may be related to an applicant's credit worthiness and default risk such as financial profiles, employment history, and past behavior.

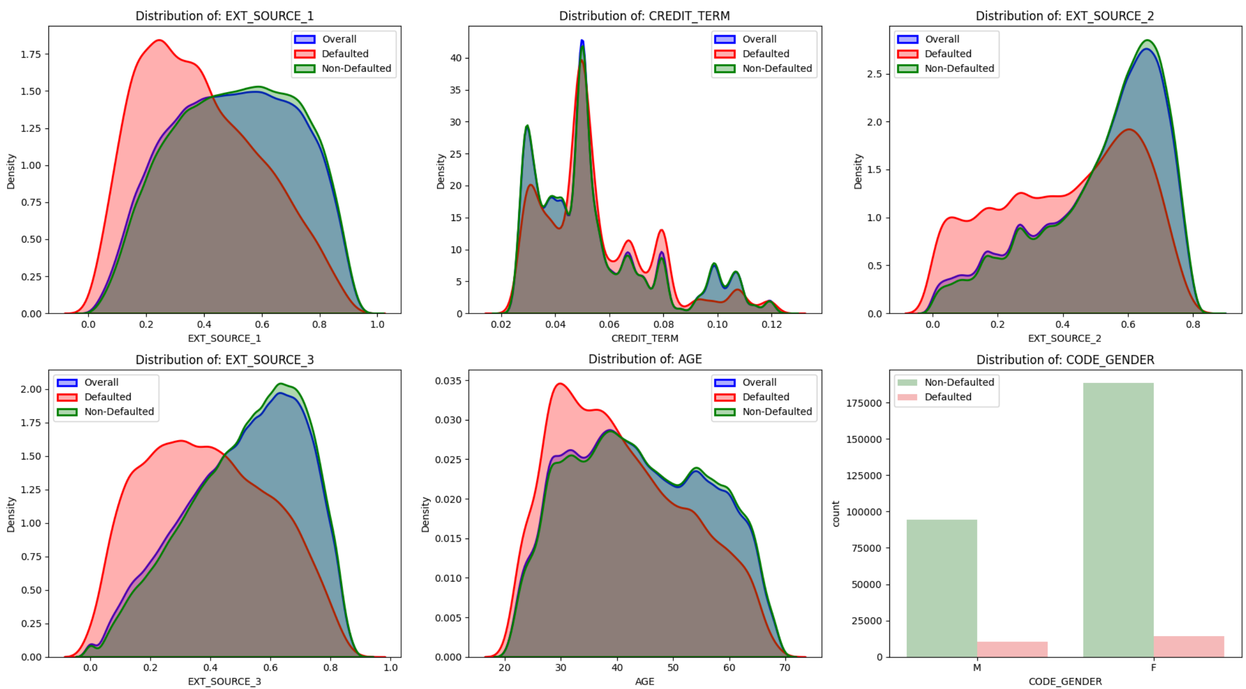

Figure 1

There are 1051 features used as inputs in the model and 307,511 entries within the data files, however, for explainability, we have decided to narrow that down to the 6 most important features – EXT_SOURCE_1 (float64), CREDIT_TERM (float64), EXT_SOURCE_3 (float64), EXT_SOURCE_2 (float64), AGE (float64), and CODE_GENDER (object). The first 5 of these features had the highest importance scores (tree-splits) of all the features within the data files. The last variable, CODE_GENDER, is important because it is our sensitive feature as we are trying to understand how sex plays a role in credit approval and default risk. Amongst the 307,511 entries, only 134,133 have a valid EXT_SOURCE_1 value, 307,499 have a valid CREDIT_TERM value, 306,851 have a valid EXT_SOURCE 2 value, 246,546 have a valid EXT_SOURCE_3 value, all 307,511 have a valid AGE value, and 307,509 have a valid CODE_GENDER value.

The EXT_SOURCE features refer to normalized scores representing an applicant's creditworthiness as calculated by third-party providers. Thus, it is unshocking that they hold such great importance relative to the other features. CREDIT_TERM refers to the duration or length of the loan. AGE is in years, and is self-explanatory – older applicants may have more extensive credit history than younger applicants. Lastly, CODE_GENDER is a binary variable representing the sex of the applicant (either male or female). It is important to note that these data files contain a large class imbalance. The dominant class is non-defaulters who account for 92% of the data. The other 8% represent defaulters. This can create challenges because the model is incentivized to predict most as non-defaulters to maximize accuracy.

By observing the distributions among these inputs (Figure 1), we can see that defaulted loans tend to have lower external scores than the distribution of loans that weren't defaulted on.

Non-defaulted loans tended to follow the overall distribution. We observe that young borrowers in their 20s to 30s tend to default on loans than older borrowers. In our sensitive feature, sex, we observe that males have a higher rate of defaulting on loans than females. By analyzing the correlation matrix of these features (Figure 2) we can see that overall the features are uncorrelated with the exception of age and the first external source score which has a positive correlation of 0.6 and the external source and sex which has a positive correlation of 0.31. These low correlations are most likely due to how the features were selected.

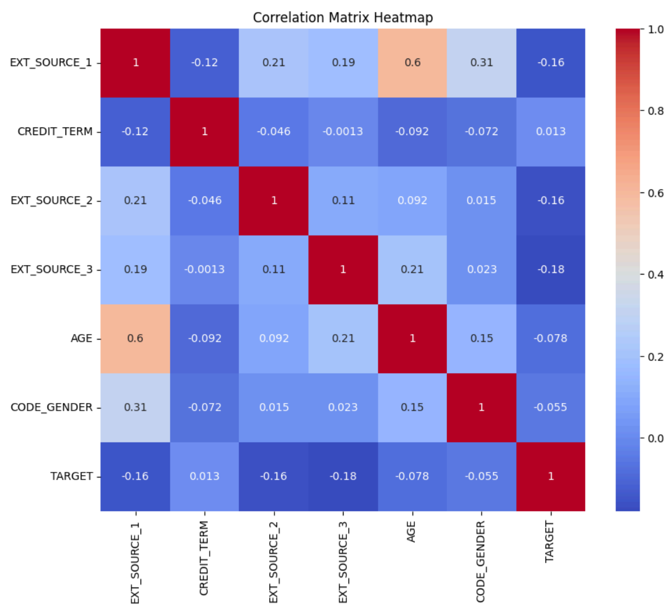

Figure 2

The output of the system is denoted as the variable called TARGET. The model provides probabilities of default, but given that this is a binary classifier, the prediction yields either 0 or 1. 0 represents a low default risk and 1 represents a high default risk. The threshold for this is set at 0.5. Meaning that anyone below this threshold would be labeled 0 and anyone above would be labeled 1.

Implementation and Validation

To implement the ADS model, the author conducts a series of data cleaning and preprocessing steps on the data provided by Home Credit Group. The author first starts by identifying anomalous values in DAYS_EMPLOYED and AMT_PAYMENT as values three standard deviations away from the mean. These values are converted to null values. The author then converts time features, which are in days, to years. They find that day features with a value of 365243 are actually null and convert them . The author creates new features like ‘CREDIT_TERM’ and ‘INCOME_BAND’ using the existing features. Then all features with over 60% null values were removed. The null values in the remaining features were filled via a custom imputer. Numerical columns were filled with median values, and categorical values were filled with the mode value. To set a baseline for performance, the features were used to train a logistic regression model which resulted in a AUC score of 0.7439. The library ‘featuretools’ was used to create a relational database among the data tables. A deep feature synthesis algorithm was run to apply basic mathematical operations across relationships to flatten the database into a 3217 feature database. Features with 60% missing values and features with only one unique value were removed from the new dataset. To remove collinear features, the correlation matrix is calculated. If a pair has an absolute correlation coefficient greater than r = 0.80, then the second feature is removed to preserve the baseline features that are ordered first. The median is then imputed for null values in numerical features and normalized, and null boolean values are converted to false. This process results in 1051 features for modeling.

To choose the proper model and meet the stated goal of the Kaggle competition, which is maximizing the AUC score, the author performed undersampling on the majority class (TARGET=0). The data was first separated into a 75:25 train/test split, and then the training set was then undersampled so that non-defaulting applicants outnumber defaulters by 2:1. This ensures that models can properly learn the differences among the two despite the major class imbalance. A series of 7 different models were then tested. The LightGBM model with early stopping performed the strongest with an AUC score of 0.781, so that model was selected for classification. To ensure the model generalized well to unseen data, a 5-fold cross validation strategy was used. To test optimization methods, the author documented AUC scores of three techniques across 50 iterations: Random Search, Grid Search, and Bayesian Optimization. Their analysis showed that grid search resulted in a consistent AUC score of about 0.77 across iterations. Random search scores were scattered with no trend, but one of its iterations resulted in the highest score. Bayesian optimization showed a clear improvement as iterations increased, however, the author noted that 50 iterations wasn't enough to show significant improvement.

Outcomes

For our analysis we decided to calculate the utility and fairness metrics of the ADS with respect to our sensitive feature sex. Overall, the model was accurate with an accuracy of 72.63% and a recall of 69.08% indicating that the model correctly identified the defaulted loans well (Figure 3). However, precision was very low at 19.23% indicating that most of the positive classifications were incorrect. Precision measures how many of the predicted defaulters actually defaulted, thus a low precision score suggests that the model is incorrectly flagging many worthy applicants. The false negative rate was 30.91% and the false positive rate was 27.05% indicating poor performance. Recall and FNR are particularly relevant within the context of credit default risk analysis, recall measures how many true defaulters were actually detected, while FNR captures the proportion missed, both metrics reflect the financial risk of approving a bad loan. On the other hand, FPR is important because it represents a missed opportunity where the lender could've lent to someone who would've repaid. It is also worth noting that the accuracy score here does not fully capture the model's behavior between the two classes primarily attributable to the large class imbalance (92% non-defaulters / 8% defaulters).

When we compare the performance among men and women we can see that the model performs differently. The model's accuracy among women was 76.56% while among men it was

65.09% (an ~11.5 percentage difference). Precision was similar among groups with 19.22% and 17.86% for women and men respectively. Recall was better for men at 76.02% and 63.89% for women suggesting that the model is quite good at detecting male defaulters but less effective in identifying female defaulters. The FNR among women was 36.10% and 23.97% among men. Men had a higher FPR at 36.13% and 22.46% for women. This indicates that men are more likely to be falsely classified to default on a loan while women are more likely to be falsely classified as safe applicants.

Looking at the fairness metrics we can see that the FPR difference and FNR difference are about the same at 13.67 and 12.12 percentage points respectively. This further backs our observations that men are more incorrectly flagged as likely to default showing that they disproportionately bear the cost of false positives. Conversely, women are more likely to be approved for bad loans showing that they disproportionately bear the cost of false negatives. The selection rate difference of 14.78 percentage points means men are more likely to be classified as defaulting on a loan than women. The demographic parity ratio of 0.63 indicates the same thing. The equalized odds ratio was low as well at 0.62 indicating that there is a major fairness issue when it comes to false selections. These fairness metrics were selected because they assess the model's ability to make equitable predictions under the group fairness framework. Demographic parity measures whether outcomes are assigned equally across groups regardless of individual characteristics. Similarly, equalized odds mandate that error rates are consistent across groups. These two metrics are both highly important in the context of credit lending, because we'd want to ensure that group membership does not influence lending decisions.

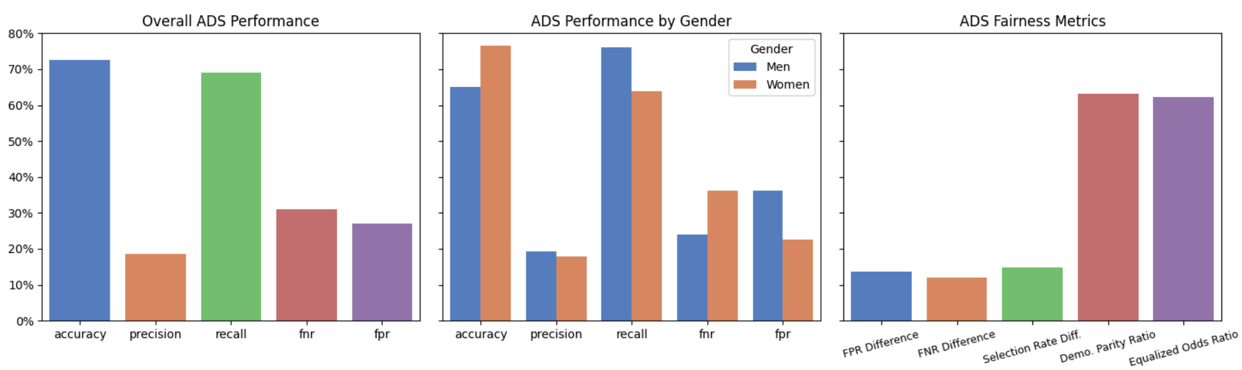

Figure 3

To promote interpretability of our ADS' behavior across sex, we assessed feature importances via SHAP values across sexes, to understand if different features drive the model's classification more for males and females. We observe that there are minor differences in the ordering of feature importance. For example, sex is the third most important feature for

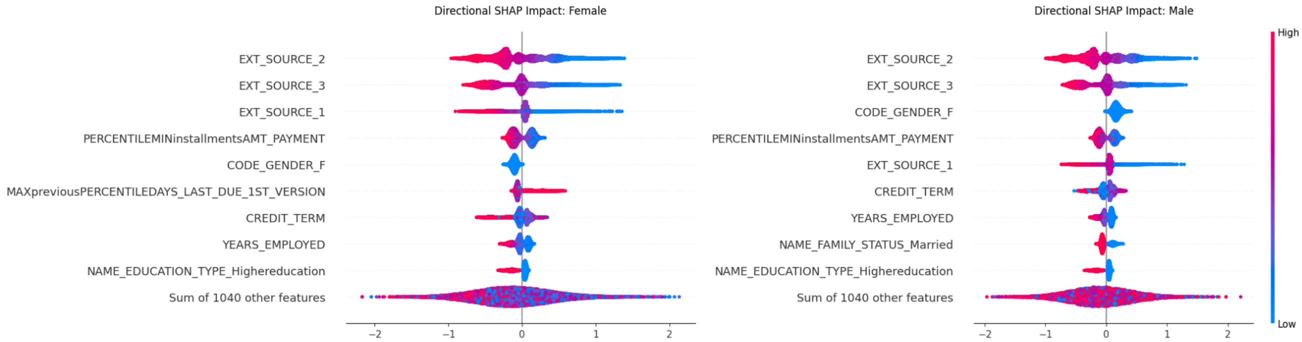

Figure 4

males, while it is the fifth most important for females. Furthermore, the plot indicates that being male drives the model towards a positive classification (being more likely to be flagged as a defaulter), while the opposite is true for females (Figure 4). Similarly, marriage status is the 8th most important feature for males, while this feature isn't among the top ten features for females. We observe that being married reduces the likelihood of defaulting. This displays an intersectional bias against unmarried men since this feature is more important for male applicants than females.

To expand on our SHAP analysis and truly understand the causal mechanism of our model we constructed a DAG of our features. Given the computational constraints of our 1051 feature model, we created a simplified DAG yet faithful of the 30 most important features globally according to our SHAP analysis. We chose to select 30 features, via the elbow method on our mean absolute SHAP values (Figure 5). From there we grouped features into logical categories to establish edges. Categories included: root nodes (age and sex), socioeconomic status, financial standing, loan details, and credit history. In the end our DAG consisted of 112 edges.

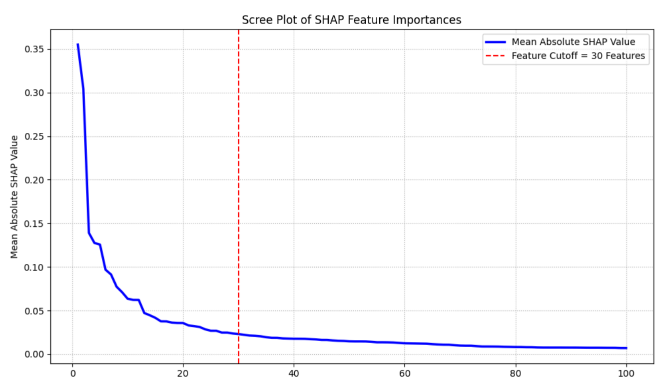

Figure 5

With the establishment of our DAG, we were able to use DoWhy to estimate the average treatment effect of being female on the model's predicted probabilities of default with a sample of our test set (n = 10,000). Our causal model estimated an ATE of -0.07973. This estimate reveals being a female causally reduces the model's predicted probability of defaulting by about 8 percentage points. To test the robustness of our estimated ATE we ran a series of causal refutation tests including using a placebo treatment (p = 0.98), adding a random common cause (p = 0.98), and using 90% subsets of our data (p = 0.98). All tests returned results in favor of the robustness of our estimate, increasing our confidence in our calculated ATE. This is not simply a correlation, but the causal effect of sex on the model output on the population level. This is a major fairness concern as the model is not sex-neutral in its decision making, but actively factors in the sex of the applicant. This can cause external biases about sexes to affect outcomes even when two applicants are identical in every other aspect. The model is inherently biased in assigning women lower default likelihood.

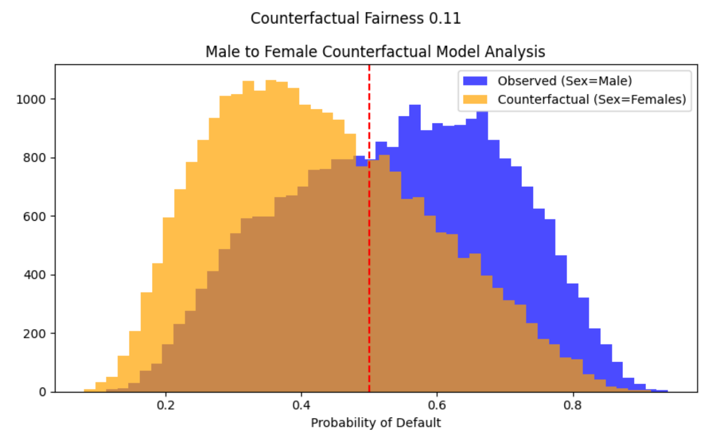

Figure 6

To build on our ATE estimate, we then evaluated counterfactual fairness. As laid out by Kusner et al. (2018), a model is counterfactually fair if flipping a sensitive feature, and adjusting the downstream causal descendants results in the same model output. By fitting a structural causal model to our DAG, we generated counterfactual cases for all males in our test set, simulating their features as if they were female. The counterfactual fairness score, which is the average change in the model's output probabilities, was roughly 0.11 displaying that the model isn't counterfactually fair, and predictions are more favorable for counterfactual females. While counterfactual fairness operates on the individual level, our findings on aggregate explain the ADS' failure to achieve group fairness. This is visually evident in Figure 6, showing the distributions of model probabilities of the observed data vs the counterfactual. We observe a shift in the distribution below the default classification threshold. This provides the causal mechanism to explain why our male population achieves a lower approval rating. However, we acknowledge that counterfactual analysis on a variable like sex is contentious, because categories like sex can exhibit "ontologically instability" where categorization can change behavior. Despite these considerations, our counterfactual fairness analysis deepens our causal understanding of the ADS.

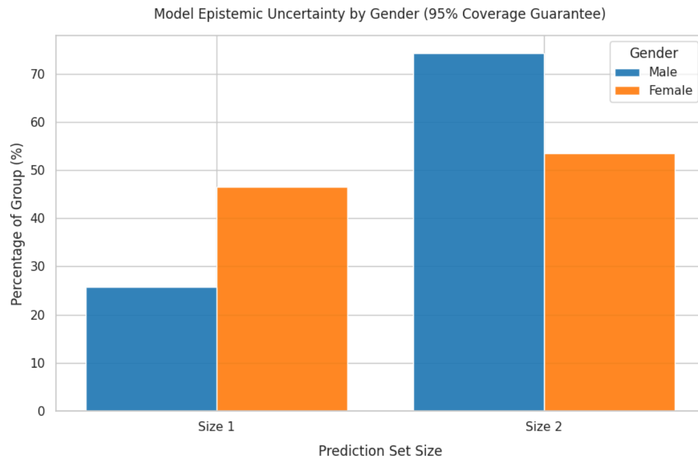

Figure 7

Finally, we empirically measure epistemic uncertainty across sexes with split conformal prediction. Understanding how epistemic uncertainty varies across genders is important to our analysis, because if the ADS is less certain for one sex compared to the other, then predictions for people of that sex will be less stable compared to those of the other one. We calculated the group specific non-conformity scores to determine the empirical quantiles for male and female applicants. This allowed us to calculate the prediction sets for each applicant. We found that the average set size for males was 1.74 while females were 1.54. Looking at the plot we see that for size 1, women make up a greater percentage of the group (~46% as opposed to ~25%). This indicates that the model is more confident in assigning women one class. On the side, for size 2, men make up a larger share of that prediction set size group (~75% as opposed to ~53%). This indicates that the model is more uncertain with assigning default or no default to males. The model is much more uncertain with male applicants than it is with female applicants. This matters in the context of fairness because the model has higher epistemic uncertainty for males rendering its predictions for that group as being less reliable and practical.

Fairness Interventions

From our analysis, we revealed several flaws in the construction of the ADS, as well as how it performs across demographics. Firstly, the model is trained on protected attributes, including sex, age, and family status. This is a clear case of disparate treatment, meaning the ADS is not legally compliant for deployment. Secondly, to mitigate the impact of the class imbalance during model training, the original author used an undersampling technique. This resulted in the training data's defaulters being primarily males, which inadvertently trained the model to use sex as a proxy for default status. Finally, we observed disparity along all five of our fairness metrics revealing a critical lack of group fairness.

To improve fairness and legal compliance, we implemented three fairness interventions aimed at addressing the issues we discovered in our baseline analysis. First we implemented the pre-processing technique of "fairness through blindness". By removing protected attributes from the model's training we eliminated inherent disparate treatment, thus improving the model's legal

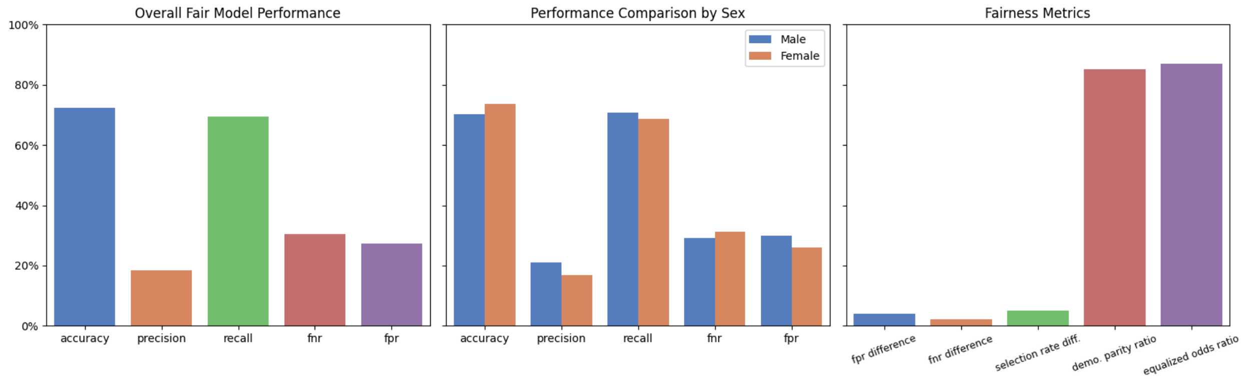

Figure 8

compliance. For our in processing intervention, we used the Exponentiated Gradient reduction algorithm optimized for Equalized Odds to improve parity in the model's error rates. We trained the model with a fairness constraint distance bound of ε = 2% to enforce this parity. Finally, as a post-processing step to account for the fairness-accuracy trade-off, we performed decision threshold tuning (threshold = 0.45) to recover lost recall performance.

These combined fairness interventions yielded promising results (Figure 8). Overall the change in raw utility metrics was negligible. Accuracy dropped by 0.21 percentage points (72.63%% to 72.42%), and precision fell by 0.8 percentage points (19.23% to 18.43%). Crucially, recall slightly improved by 0.47 percentage points (69.08% to 69.55%), the false negative rate dropped by 0.44 percentage points (30.91% to 30.45%), and the false positive rate increased by 0.28 percentage points (27.05% to 27.33%). Furthermore, the disparity in utility metrics between demographics shrank significantly, particularly for accuracy, recall, FNR, and FPR.

The fairness metrics showed major improvements across the board. The FNR difference reduced to 2.13 percentage points from 12.12. Similarly, the FPR difference reduced to 3.93 percentage points from 13.67. The Selection Rate Difference reduced from 14.78 to 5.05 percentage points, and the Demographic Parity Ratio increased from 0.63 to .85. Combined, these metrics demonstrate that the model predicts defaults for both males and female applicants at similar rates post-intervention. Finally, the Equalized Odds Ratio increased massively from 0.62 to 0.87 indicating a robust mitigation of the model's baseline disparate error rates. While our fairness interventions successfully improved our model's performance along the metrics of interest, it is important to note that given the Fairness Impossibility Theorem our improvements mathematically necessitated a trade-off, coming at the expense of prediction parity across demographics.

To see how sex causally affects our mitigated model's predictions, we estimated the average treatment effect of being a female applicant on the model's predictions post-intervention. We estimated an ATE of -0.00794, meaning that, on average, being female decreases the mitigated model predicted probability of default by less than 1 percentage point. This demonstrates that our fairness interventions essentially eliminated the causal relationship between sex and the model's final decisions. We then ran the same three robustness refutation tests: placebo treatment (p = 0.88), random common cause (p = 0.94), and 90% subset (p = 0.66). All refutation tests were statistically insignificant, confirming the robustness of our estimate. This marks a substantial improvement from the model's baseline ATE of -0.07973. Allowing us to confidently state that the final model is causally fair.

Summary

We believe that the data was appropriate for this ADS because it contained extensive information pertaining to applicants including their financial profiles, employment history, and credit behavior. All of this information is highly relevant in assessing default risk. Home Credit Group, which collected and supplied this data, utilizes these features when evaluating loan applications. Therefore, we know we are working with domain relevant information. However, as with most data, there were some concerns and limitations. The biggest of which was the class imbalance (92% non-default vs. 8% default).

From our audit, we believe that the ADS is robust. The author of the ADS took many steps to find the best model for this use case and ensure that the model wouldn't overfit on the data. This included comparing the performance of seven different models to select the strongest one. Early stopping and K-fold cross validation were also implemented. These methods yielded positive results confirming that the model would perform well on unseen data. Given our accuracy metrics, we believe that the model is moderately accurate. The accuracy of our model was high at 72.63%. We chose accuracy because it provides a high level view of how well the model performs. The stakeholder that would benefit the most from accuracy is Home Credit Group, since a highly accurate model means it would be better at classifying applicants and help inform the lender's decisions. The precision of the model was very low at 19.23% and FPR was 27.05%. We believe that Home Credit Group and denied applicants would be interested in these metrics, because a low precision and a high FPR means that Home Credit Group is missing out on loans that were actually safe to approve. Similarly, a denied applicant would be heavily interested in precision and FPR, because poor performance along these metrics could mean that the applicant was denied by the model when they were truly a safe borrower, and therefore don't get the loan they need. The recall of the model was strong at 69.08%, however, the FNR was high at 30.91%. We believe that these are the two strongest utility metrics for this situation, because failure to identify a defaulter would have severe financial implications for both the lender and borrower. For these metrics we believe that the relevant stakeholders would be Home Credit Group and riskier loan applicants.

When analyzing fairness along our sensitive feature, sex, we found that the model fails to perform equitably. The FPR difference was 13.67 percentage points, with males having a high FPR. The stakeholders that this metric pertains to are male applicants, because we found that they are more likely to be falsely classified as likely to default. On the other hand, the FNR difference was 12.12 percentage points with females having a higher FNR. The relevant stakeholders here are high risk of default females, because they are more likely to get their applications approved when they shouldn't be putting them into debt that they aren't ready for and will have severe consequences on their credit. We included FPR and FNR differences, because it helps us understand how the types of errors are distributed across sex. The selection rate difference of 14.78 percentage points means males are more likely to be classified as defaulting on a loan than females. The relevant stakeholders here are male applicants, because as a demographic they are selected as likely to default, so improving this metric would improve male applicants' situations. The demographic parity ratio of 0.63 and the equalized odds ratio of 0.62 reveal that there is a major fairness issue when it comes to false selections. A relevant stakeholder here could be regulators that aim to ensure fairness among males and females. Different error rates mean that the model is performing differently across the two groups which a regulator would want to mitigate when acting from a group fairness perspective.

Based on what we have observed we would be hesitant to deploy the baseline ADS into underwriting. Firstly, the baseline ADS reveals a clear case of disparate treatment where sex, age, and marital status are included as model inputs. Using these protected classes would make the model illegal to use in practice. Beyond this, the model exhibits a clear bias against male applicants as indicated by the auditing methods that we employed. The most revealing insight came from our causal analysis which revealed sex had a strong causal effect on the model predictions. The counterfactual fairness plot further displays this bias by showing that the two distributions do not overlap. The model would've been considered fair if swapping an applicant's sex had no impact on the applicant's outcome, however, that is not the case. The last auditing method that we used was split conformal prediction. By analyzing prediction sets we found that the model was epistemically more uncertain when it came to predictions for male. This was noted by the average set size for males being higher at 1.74 than that of females which was 1.54. Clearly, the baseline model has severe legal and ethical limitations.

To transform this undeployable model into a production-ready system, we executed a rigorous fairness intervention pipeline. By enforcing "fairness through blindness" we eliminated disparate treatment and ensured legal compliance. By applying the Fairlearn Exponentiated Gradient algorithm and lowering the decision threshold to 0.45, we successfully neutralized the disparate impact while recovering the model's predictive power. Ultimately, these interventions raised our Equalized Odds Ratio to 0.87, and eliminated the causal effect of sex on model predictions. Through our fairness interventions, we successfully engineered an ADS that is not only equitable, but also accurate and legally defensible.